WEEK 5 · DISCOVER PHASE · THE GATE

User Validation Gate

The checkpoint that ends DISCOVER. Prove there's a **real problem** for **real users** — then, and only then, you're allowed to build. **Don't pass, don't start BUILD.**

Pass / Fail · 10 of 50 in Project

Reviewed: proposal · backlog · design · compliance

Dr. Prasara Jakkaew · ADT MFU

RECAP · THE DISCOVER MONTH

Everything so far was to earn one stamp

W1 · Charter → W2 · Ethics + law → W3 · Backlog → W4 · Design + 4 diagrams → **W5 · GATE** → W6 · BUILD begins

YOU'VE BEEN ASSEMBLING

- Evidence of a **real user need** (interviews / research / advisor)
- A **product backlog** that traces to that pain
- **Design** + the **4 required diagrams**
- A **rule.md** for legal / PDPA compliance

TODAY

You present that package at the gate.

A gate is not a grade for effort. It's a **go / no-go decision**: is there enough proof to justify spending September building?

WHY THIS EXISTS

Building the wrong software costs the most

WITHOUT A GATE

- Team commits to an **idea**, not to a user need
- Builds for a month on a **guess**
- Ships something **nobody asked for**
- Discovers the real problem in November — too late

WITH A GATE

- You **prove problem–solution fit first**
- Cost of being wrong = **a few interviews**, not a month
- BUILD starts with **evidence**, not opinion
- This is exactly how industry stage-gates work

💡 **IN PLAIN TERMS** Failing here, on paper, costs far less than failing in December with a finished product no one wants.

THE GATE CHECKS FOUR THINGS

Four pillars — you need all four

Pillar 1

Validated need

Interviews, user research, or advisor brief — **with evidence**. + usability test planned.

Pillar 2

4 diagrams

Present **and consistent** with each other + the backlog.

Pillar 3

Traceability

Every feature traces **back to a real pain**.

Pillar 4

Compliance

`rule.md` + legal requirements from W2 are in.

THE STANDARD This is **pass / fail**, worth **10 of the 50 project points**. A missing pillar doesn't cost "a few marks" — it stops BUILD until you fix it.

PILLAR 1 · VALIDATED USER NEED

Prove the need is real — three paths count

Path A · Interviews

Talk to real users

≥ 5 (target ≥ 15). Dated log + a **direct quote** of the pain; past behaviour, not "would you use…?".

Path B · User research

Competitors + community

Competitor analysis, app-store reviews, forum / Pantip / social threads — **screenshot with source + date**.

Path C · Advisor brief

Requirements from อาจารย์

Reqs collected from a faculty advisor / sponsor acting as the **client** → exempt from the interview count.

⚠ NON-NEGOTIABLE · EVERY PATH You still commit to **usability testing** with real users on your prototype / MVP (run in the TEST phase). No path skips real-user validation — it's just **deferred to the usability test**. Show a plan: who, what tasks, when.

Still doesn't count: a claim with **no evidence behind it**, or "users" who are your own teammates. Whichever path — **bring the proof**.

PILLAR 2 · THE 4 REQUIRED DIAGRAMS

Present — and consistent

D1

System Context

The system as one box + the actors and systems around it.

D2

Use Case / Story Map

What each actor does — traces to backlog items.

D3

High-level Architecture

The big components + how they connect.

D4

Activity / Sequence

One core workflow, step by step.

CONSISTENT MEANS

- An actor in D1 appears in D2
- A use case in D2 has a backlog item
- A component in D3 does the work in D4

INCONSISTENT = FAIL

- D2 shows a feature that's **not in the backlog**
- D4 calls a component **not drawn in D3**
- Four pretty pictures that **don't agree**

PILLAR 3 · TRACEABILITY

Every feature traces back to a real pain

Pain (a user said it) → **Requirement** → **Backlog item** → **Design / diagram**

PAIN (USER'S WORDS)	REQUIREMENT	BACKLOG ITEM	SHOWS UP IN
"I lose the paper tally and re-count cups."	Log a cup in ≤ 3 taps, stored safely	`US-01 quick cup log`	D2 use case · D4 activity
"I never know today's total till I add it up."	Show a live running total	`US-04 daily total`	D1 context · D3 component

THE TEST WE RUN Pick **any** feature and ask **"which user asked for this?"** No answer → it is scope creep, and it gets cut, not built.

PILLAR 4 · COMPLIANCE REVIEW

Your W2 law homework becomes a gate check

AT THE GATE, WE CHECK YOUR RULE.MD FOR

- **PDPA** — if you store personal data, is consent + purpose declared?
- **Computer Crime Act §26** — are logs kept ≥ 90 days?
- **Electronic Transactions Act** — any e-signature / e-record handled right?
- Are these written as **real backlog requirements**, not just prose?

WHY NOW, NOT LATER

Compliance is a design input, not something you add later.

"We'll add PDPA later" means rebuilding your data model in October. We catch it here while it's still one line in `rule.md`.

LINK TO W2 The `rule.md` your company wrote in the Week 2 workshop is the file reviewed here — it should sit at repo root next to `CLAUDE.md`.

HOW YOU'RE JUDGED

Three outcomes — PASS, CONDITIONAL, FAIL

✓ PASS

All four pillars stand. Diagrams present **and** consistent, need validated (any path), traceability holds, compliance in. **BUILD unlocked.**

◐ CONDITIONAL

One pillar has a fixable gap (e.g. evidence thin, no usability-test plan, one diagram thin). Fix it this week, re-submit. BUILD starts once cleared.

✕ FAIL

Diagrams **inconsistent**, no real user evidence, or no traceability. Rework DISCOVER — **no BUILD** yet.

💡 IN PLAIN TERMS PASS = you can start BUILD. CONDITIONAL = one small fix needed. FAIL = the design is not ready yet. **Most gaps are CONDITIONAL, and fixable in days.**

HOW TODAY RUNS · 180 MIN

Teach → self-check → peer review → gate stamp

1 0–30 MIN

The gate, explained

These slides: the four pillars, the rubric, how to present. **You're here now.**

2 30–45 MIN

Team self-check

Each company runs the **Student Gate Checklist** on its own package. Mark every pillar green / amber / red. Be honest — the peer round will find it anyway.

3 45–110 MIN

Peer mini-reviews (rotating)

Companies pair up. Each gives a **5-min gate pitch**, the other scores it with the scorecard and names **one PASS-blocker**. Then swap. Two rounds.

4 110–165 MIN

Instructor gate stamp

I visit each company for a short review and record PASS / CONDITIONAL / FAIL on the scorecard. Fix time runs in parallel.

5 165–180 MIN

Close + fix-it plans

CONDITIONAL teams write their fix-it plan and re-submit date. Everyone previews W6 BUILD.

YOUR 5-MINUTE GATE PITCH

Say these six things — in this order

1 · 30s

The problem

One sentence. Whose pain, and how you know it's real.

2 · 60s

The evidence

Your path — interviews / research / advisor — with **one concrete proof** + your usability-test plan.

3 · 60s

The trace

Show **one** pain → requirement → backlog → diagram.

4 · 90s

The 4 diagrams

Show them; say one consistency link between two of them.

5 · 30s

Compliance

One line from `rule.md` that's now a backlog requirement.

6 · 30s

The ask

"We believe we PASS because…" — name your weakest pillar too.

TONE Not a sales pitch. You are an engineer showing **evidence** to a reviewer. Admitting your weakest pillar earns more trust than hiding it.

PEER MINI-REVIEW · HOW TO REVIEW ANOTHER TEAM

Give tough, fair feedback

GOOD REVIEWING

- Score against the **scorecard**, pillar by pillar
- Ask **"which user asked for this?"** on a random feature
- Name exactly **one PASS-blocker** — the thing to fix first
- Point at a **diagram inconsistency** if you see one

WEAK REVIEWING

- "Looks good 👍" — helps no one
- Redesigning their product for them
- Nit-picking colours instead of checking evidence
- Criticising the people instead of the work

HAND BACK A filled scorecard + one sentence: **"Your PASS-blocker is ___; fix that first."** Score it on the **live scorecard** → prascymj.github.io/w5-gate-scorecard-2569

PEER MINI-REVIEW · THE ROTATION, STEP BY STEP

Two rounds → two different reviewers each

1 0–5 MIN

Team 1 pitches

The 5-min gate pitch. Team 2 just listens with the scorecard open.

2 5–8 MIN

Team 2 scores Team 1

Mark P / C / F on all four pillars in the **live scorecard**, then name **one** PASS-blocker out loud.

3 8–13 MIN

Swap — Team 2 pitches

Now roles flip. Same 5 minutes, Team 1 listens.

4 13–16 MIN

Team 1 scores Team 2

Same drill. **One round ≈ 16 min** — both teams reviewed once.

ROUND	WHO PAIRS WITH WHO (↔ = PITCH, THEN SWAP & SCORE)	TIME
Round 1 pair up	A ↔ B · C ↔ D · E ↔ F	~16 min
Shuffle	Everyone moves to a **new** partner (no repeats)	2 min
Round 2 new partner	A ↔ D · B ↔ F · C ↔ E	~16 min

RESULT Every team **pitched twice**, **reviewed two other teams**, and walks away holding **two scorecards + two PASS-blockers** — so you get feedback from two teams, not just one. The rest of the 45–110 block is **fix time**.

ODD NUMBER OF TEAMS? Make **one trio** instead of a pair — it round-robins (3 pitch, 3 score) and takes one extra ~8-min slot. Adapt the A–F letters to however many companies you have today.

THE GATE SCORECARD · WHAT I MARK

The same sheet peers and I use

PILLAR	CHECK	✓ / ◐ / ✕
1 · Validated need	Interviews (≥5) / user research / advisor brief · evidence shown · usability test planned	P / C / F
2 · 4 diagrams	D1–D4 all present · consistent with each other + backlog	P / C / F
3 · Traceability	Every backlog item traces to a pain · no orphan features	P / C / F
4 · Compliance	`rule.md` present · PDPA/CCA/ETA as real requirements	P / C / F

DECISION RULE **All P → PASS.** One C, rest P → **CONDITIONAL.** Any F, or two+ C → **FAIL**. Notes + fix-it date go on the same sheet.

📱 Score online — open on your phone in class:

prascymj.github.io/w5-gate-scorecard-2569

Open the live scorecard →

THE FIVE WAYS TEAMS MISS THE GATE

Know them, avoid them

THE MISTAKE	WHAT WE CHECK	THE FIX
No real evidence	A claim with no log, screenshots, or advisor brief — or no usability-test plan	Show one path: interviews, captured research, or advisor reqs — + plan the usability test
Solution-first	A cool feature no user ever asked for	Cut it, or find the pain it answers
Diagrams disagree	D4 uses a component not in D3; D2 ≠ backlog	Make one edit ripple to all four + backlog
Compliance as prose	"We care about privacy" but no requirement	Turn each `rule.md` rule into a backlog item
Building already	Production code before the gate	Stop. DISCOVER has no code. That's the point.

IF YOU GET CONDITIONAL

It's a to-do list, not a verdict

YOUR FIX-IT PLAN (WRITE IT TODAY)

- **The one gap** — the pillar that came up amber
- **The concrete fix** — e.g. "interview 2 more users", "redraw D3 to match D4"
- **Who + by when** — an owner and a re-submit date

THEN

Re-submit → cleared → BUILD.

Most teams are CONDITIONAL, not FAIL. A gap found here costs you a couple of days — the same gap found in October costs you the project.

DEADLINE Fix-it plans are due at close today; re-submission before **W6 (Sep 16)** so scope-lock starts on time.

WHAT PASS LETS YOU DO

PASS unlocks BUILD — and the rules change

GATE · PASS → **W6** Scope lock (MoSCoW) + Sprint 0 + repo + MCP →

W7–8 Build sprint → Alpha Demo

NOW ALLOWED

- **Production code** — full AI / vibe-coding
- Ship **one core workflow** a real user can finish

STILL NOT ALLOWED

- **Feature creep** — scope is locked at W6
- Rewriting the whole thing chasing a new idea

MINDSET SHIFT DISCOVER asked **"are we building the right thing?"** BUILD asks **"are we building the thing right?"** The gate separates the two.

THE MODEL · A GATE-READY PACKAGE

What "PASS" looks like — CupLog

Pillar 1 · need

7 baristas interviewed across 3 cafés (Path A) + app-store reviews captured. Shared pain: **paper tally gets lost**. Usability test planned for TEST.

Pillar 2 · diagrams

D1–D4 present. The *Barista* actor in D1 does *Log cup* in D2, which D4 walks step-by-step.

Pillar 3 · trace

Every backlog item (`US-01…US-06`) links to a quoted pain. No orphan features.

Pillar 4 · compliance

`rule.md` at root: PDPA consent for staff names, CCA §26 90-day logs — each a backlog requirement.

WHERE TO STUDY IT Full worked example in **CupLog-Worked-Example-2569** (§6 diagrams, §2 `CLAUDE.md`+`rule.md`). Copy its *structure*, not its content.

ACTIONS · BY THE END OF TODAY

Leave class with a stamp or a plan

EVERY COMPANY

- Run the **Student Gate Checklist** on your package
- Give + receive a **5-min peer gate pitch**
- Get your **instructor gate stamp** recorded
- Submit the DISCOVER package to Google Classroom

IF CONDITIONAL / FAIL

- Write your **fix-it plan** (gap · fix · owner · date)
- Re-submit before **W6 · Sep 16**
- No BUILD until cleared — protect your own timeline

REMINDER The gate is graded **inside the Project (10/50)** and is **separate** from the written Midterm. It's about evidence, not memorised facts.

WEEK 5 · CLOSE

Prove it, then build it.

A gate feels strict today. By December it will be the reason your team **stopped guessing**. Real users, four diagrams that agree, features that trace to a real user need, and compliance built in — **that** is what passes.

Next · W6 (Sep 16): Scope lock · Sprint 0 · repo · MCP

Don't pass, don't start BUILD.